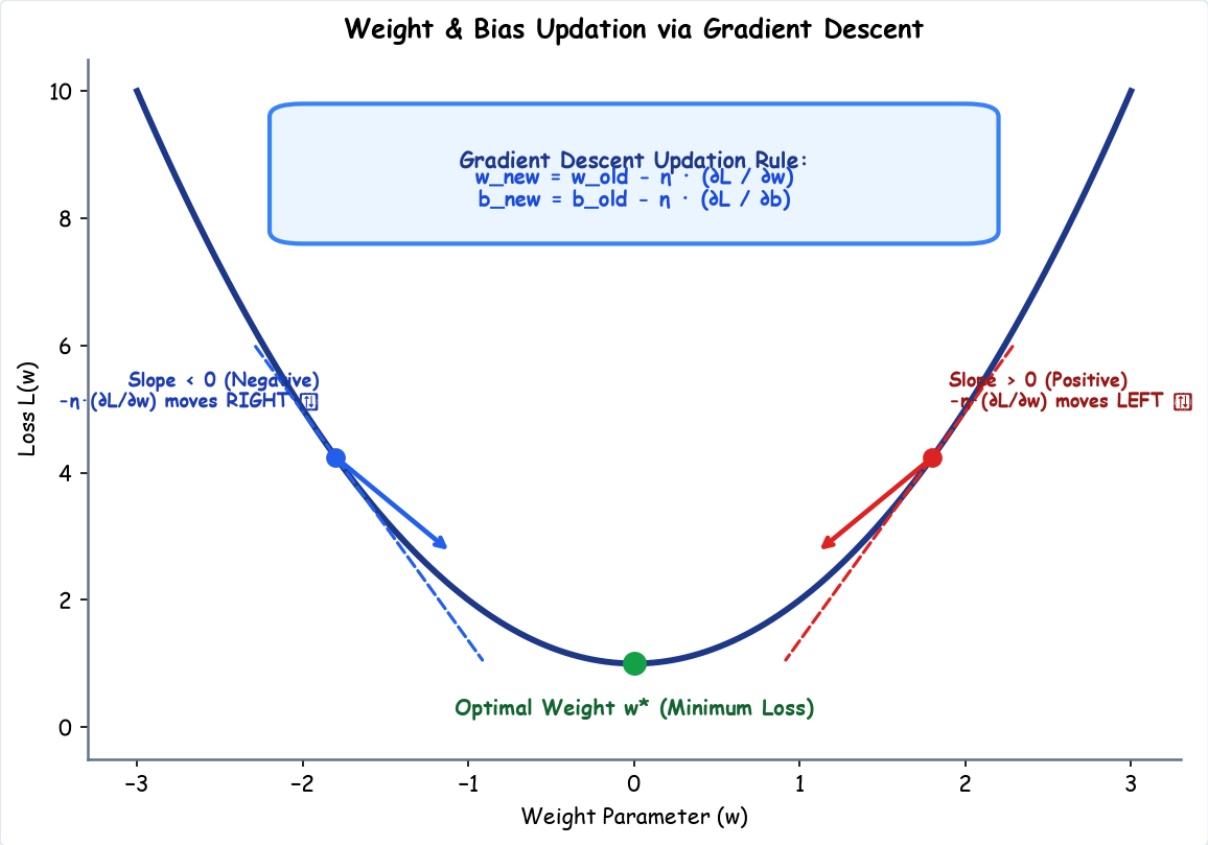
Submodule 37.04: Weight & Bias Updation (Hinglish)

1. Weights aur Biases Update Kaise Hote Hain?

Jab model ka loss nikal jata hai, tab network ke andar ke **Weights (W)** aur **Biases (b)** ko Gradient Descent ke zariye update kiya jata hai taaki agle round me error kam ho sake.



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2. Updation ka Golden Formula

$$w_{new} = w_{old} - \eta \cdot \frac{\partial \mathcal{L}}{\partial w}$$

$$b_{new} = b_{old} - \eta \cdot \frac{\partial \mathcal{L}}{\partial b}$$

- w_{old}: Purana weight.
- η **(Learning Rate)**: Step ka size tay karta hai ki kitni tezi se aage badhna hai.
- $\frac{\partial \mathcal{L}}{\partial w}$: Curve ka slope (gradient).

- **Minus ($-$) sign**: Hamesha slope ke ulti direction me chalkar minima tak pahunchata hai.

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3. Mathematical Intuition: Yeh Formula Hamesha Kaam Kyun Karta Hai?

1. **Agar hum minimum ke Right side khade hain (Positive Slope)**:
    - Slope positive hai (> 0).
    - Formula: $w_{new} = w_{old} - (\text{positive})$.
    - Weight kam hoga aur **Left ki taraf** minimum ki ore shift hoga!
2. **Agar hum minimum ke Left side khade hain (Negative Slope)**:
    - Slope negative hai (< 0).
    - Formula: $w_{new} = w_{old} - (-\text{value}) = w_{old} + \text{value}$.
    - Weight badhega aur **Right ki taraf** minimum ki ore shift hoga!

Dono hi suraton me yeh formula points ko kheench kar minimum tak le aata hai!\n